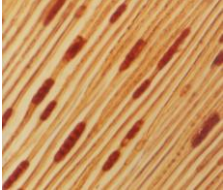
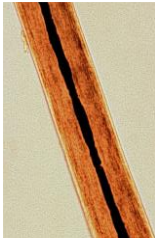


Trace Evidence

- Fibers
- Hairs
- Seeds
- Feathers
- Soil
- Glass
- Building Materials
- Paint Chips



Hair and fiber analysis by microscopic comparisons



Locard's Exchange Principle



Locard's Principle Of Exchange

- Edmund Locard was a French forensic microscopist. He developed the Locard Principle of Exchange.
- The Locard Exchange Principle is the foundation of trace evidence.
- This principle states that whenever two people or objects come into physical contact an exchange of materials takes place.
- These materials may have significance in a criminal investigation
 - fibers from clothing, carpet, blankets, etc.
 - hairs from different body areas
 - chips of nail polish
 - automotive paint
 - broken glass



Primary and Secondary Transfer

Primary Transfer – also called direct transfer

Example: head hair falls from your head onto your jacket

Secondary Transfer – also called indirect transfer.

Example: A head hair falls from a person's head into a chair. The next person who sits in chair picks up the head hair. The two individuals never had direct contact with each other.



Collection

- Scraping Technique
- Vacuum



Processing Room for Trace Evidence

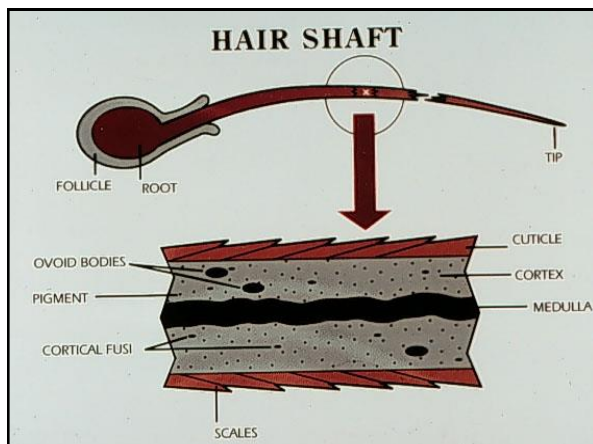
- Cleaned before each use
- Suspect, victim and crime scene samples processed in different rooms to avoid contamination
- PPE is worn and changed before entering a separate room



Hair Evidence

- Animal vs. human
- Racial Characteristics
- Body Area
- Damage/disease
- Artificial Treatment
- Microscopic characteristics
- Forcibly removed or naturally shed





Hair Structure

Cuticle - a thin translucent layer surrounding the hair shaft (scales)

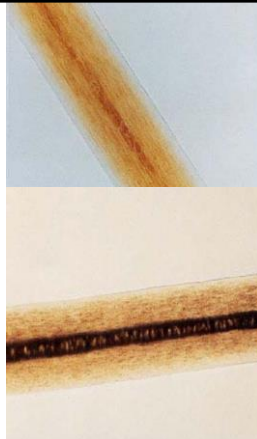
Medulla - a cellular column running through the center of the cortex

Cortex - The cortex is the main body of the hair composed of elongated and fusiform (spindle shaped) cells. It may contain cortical fusi, pigment granules, and/or large oval-to-round-shape structures called ovoid bodies.

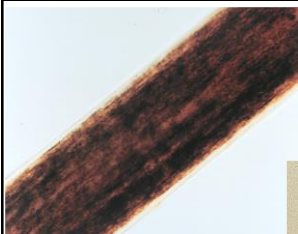
Human or Animal

In humans, pigment granules are commonly distributed toward the cuticle. Scales have an imbricate pattern.

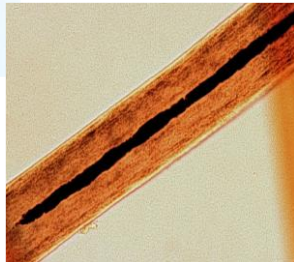
Animal hairs have the pigment granules commonly distributed toward the medulla. Scales differ depending on animal.



Racial Origin



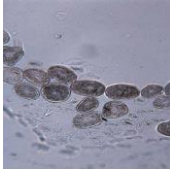
Racial characteristics can be microscopically determined.



Identification of Human Hair

Racial Origin: Caucasian - European

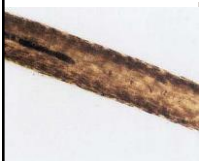
- Shaft diameter: minimal variation
- Pigment granules: sparse to moderately dense with even distribution
- Cross-sectional shape: oval/elliptical



Identification of Human Hair

• **Racial Origin:** Negroid – African

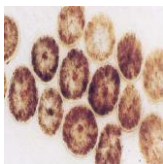
- Shaft diameter: considerable variation
- Pigment granules: densely distributed and arranged in prominent clumps
- Shaft: prominent twist and curl
- Cross-sectional shape: flattened/elliptical



Identification of Human Hair

• **Racial Origin:** Mongoloid - Eastern Asian, Eskimo-Aleutian, and Native American

- Shaft diameter: coarse and usually with little or no variation
- Pigment granules: densely distributed and often arranged in large patchy areas or streaks
- Medulla: prominent (often broad and continuous)
- Cuticle: thick
- Cross-sectional shape: round



	EAST ASIAN	CAUCASIAN	AFRICAN-AMERICAN
HAIR CROSS-SECTION			
AVERAGE STRAND COUNT	500,000 PER PERSON	700,000 PER PERSON	200,000 PER PERSON
GENERAL STRAND SHAPE			





Body Area

Hair buckled = pubic hair



Triangular hair = facial hair

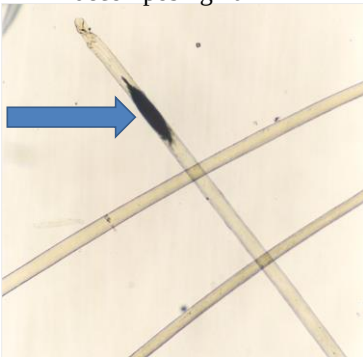
Limb hairs have minimal variation – little to no value

Damage or Disease



Lice or egg sacs

“Dead man’s ring” – typical characteristic of decomposing hair



Growth of Hair

Hair follicles go through three (3) distinct phases:

- Anagen - active growth phase; human head hairs spend about two to four months in this phase; about 80-90%
- Catagen - breakdown phase; lasts two to four weeks, 2%
- Telogen - lasts two to four months, until they fall out naturally or are forcibly removed, about 8-18%

Forcibly removed hairs exhibit stretching and contain a follicular tag



Telogen – naturally shed



Anagen – forcibly removed

Artificial Treatment

Distinct color change
Clear line of demarcation
Oxidizes pigment granules



Growth rate = ½ inch per month

Comparative Microscopy

- Skillful microscopic technique provides the forensic hair examiner with a highly discriminating means to examine and compare hair and utilizes various types of microscopes.
- Known head hair exemplars = 25 hairs collected from various areas of the scalp



Comparative Microscopy

- **Stereomicroscope** - permits gross observations of the hairs; allows a rapid overview of the range of characteristics.
- **Cross Polarizing microscope** - delineates the hair's finer structural characteristics.
- **Transmitted light microscope** - allows a side-by-side comparison of the known and questioned hairs. Hairs are compared from their root end to their tip end.

Conclusion and Report Writing

Forensic laboratory reports of hair comparisons normally will provide one of three answers:

- The hairs are similar in microscopic characteristics, indicating an inclusion (included).
- The microscopic features of the hairs are not alike and therefore did not come from the same person (excluded).
- No conclusion can be drawn from the evidence. This may be due to insufficient specimen or unsuitable exemplars.

Fibers

Natural Fibers

Plant: cotton, jute, hemp

Animal: wool, silk

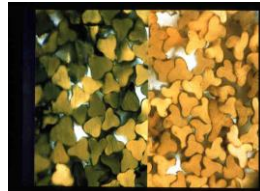
Mineral: asbestos

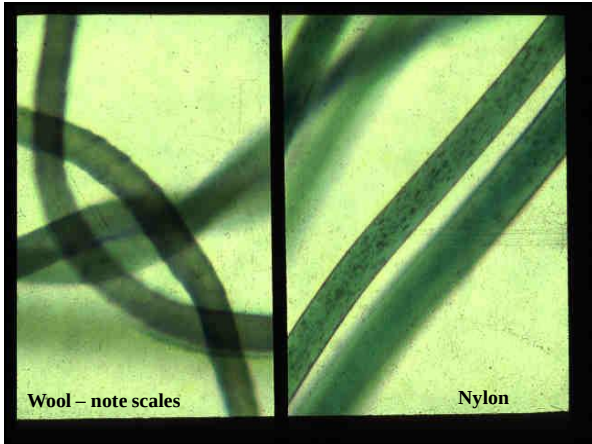
Synthetic Fibers

Nylon Rayon

Polyester Acetate

Olefin Acrylic

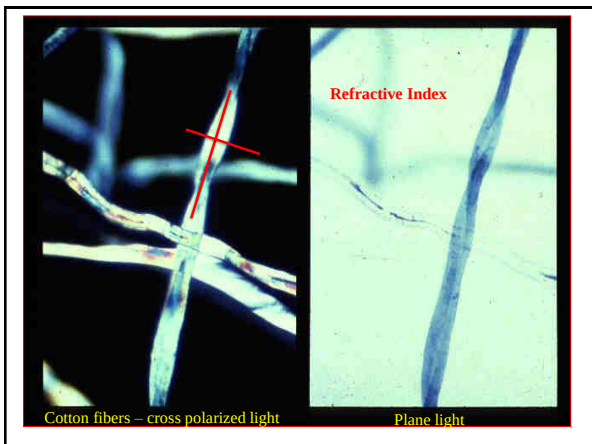




Fiber Analysis

Comparison of:

- Color –
microspectrophotometer
- Diameter
- Shape
- Presence of scales
- Chemical Composition –
FTIR and Cross Polarized
Light Microscope



Conclusion and Report Writing

Forensic laboratory reports of fiber comparisons normally will provide one of three answers:

- The fibers could have originated from the same source or the fibers are consistent with originating from the same source (included).
- The fiber features (color, diameter, shape, composition) are not alike and therefore did not come from the same source (excluded).
- No conclusion can be drawn from the evidence. This may be due to an insufficient specimen or unsuitable exemplars.

Comparison vs. Identification in Fabric Exams

Physical Matches = identification

Color, Construction (diameter and shape), and
Chemical Composition = comparison

